

Challenge (Habanero)

- Q1.** Solve the system: $x^2 + y^2 = 25$, $y = x^2 - 4$. What is the value of x ?
(A) -3
(B) -2
(C) 1
(D) 2
(E) 3
- Q2.** A chef's recipe for a circular cake requires a 16-inch diameter. What is the x -coordinate of the point on the circle $x^2 + y^2 = 64$ that is farthest from the line $x = 2$?
(A) -8
(B) -4
(C) 0
(D) 4
(E) 8
- Q3.** Solve the system: $y = x^2 - 3x + 2$, $y = -x^2 + 5x - 2$. What is the value of y ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q4.** A construction worker needs to build a rectangular foundation with an area of 48 square meters. If the length of the foundation is x meters, what is the width in terms of x ?
(A) $\frac{48}{x}$
(B) $\frac{24}{x}$
(C) $\frac{12}{x}$
(D) $\frac{6}{x}$
(E) $\frac{3}{x}$
- Q5.** The graph of the system $x^2 + y^2 = 9$, $y = x^2 - 1$ has how many solutions?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q6.** A quadratic equation for the height of a projectile is $h = -x^2 + 4x + 2$. If the maximum height is 6, what is the value of x at that height?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q7.** A chef's special sauce requires a mixture of two ingredients, x and y , where $x^2 + y^2 = 16$ and $y = x^2 - 4$. What is the value of y ?
(A) -4
(B) -2
(C) 0
(D) 2
(E) 4
- Q8.** A construction worker needs to build a triangular roof with a base of x meters and a height of y meters. The equation for the roof is $x^2 + y^2 = 25$. If the base is 3 meters, what is the height?
(A) 4
(B) $\sqrt{16}$
(C) $\sqrt{20}$
(D) $\sqrt{24}$
(E) $\sqrt{30}$

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $x^2 + y^2 = 20$, $y = x^2 - 2x + 1$. Show all work and solutions.
- Q10.** A circular fountain has a diameter of 12 inches. If the equation of the circle is $x^2 + y^2 = 36$, find the x - and y -coordinates of the point on the circle that is farthest from the line $y = x$. Show all work and solutions.

Chef's Tips

- Tip 1: Use the quadratic formula when necessary.
- Tip 2: Check for extraneous solutions.
- Tip 3: Use graphing to verify solutions when possible.